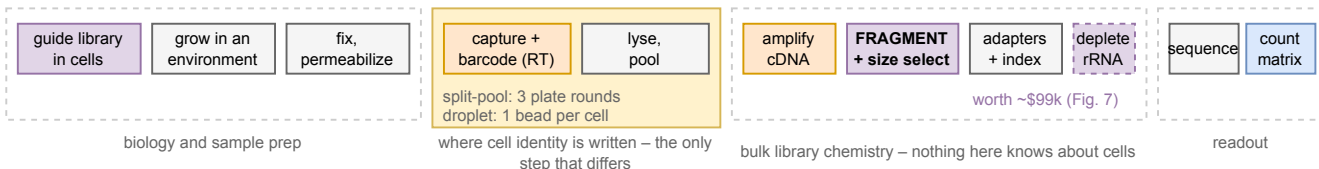


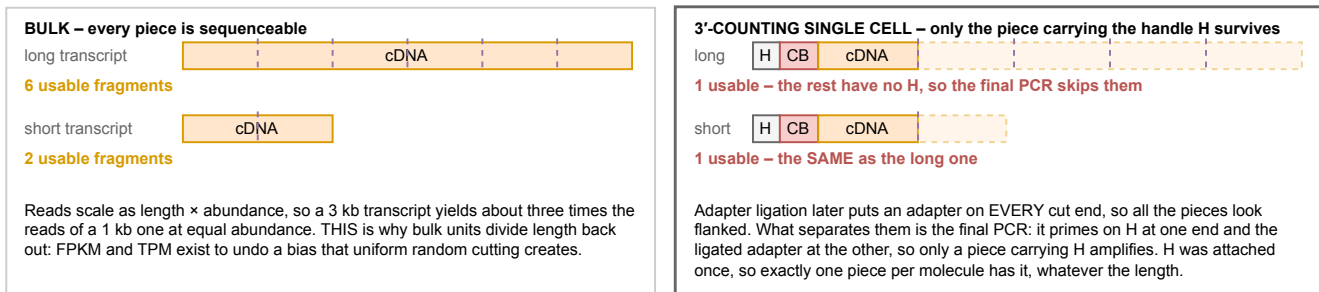
The whole experiment, and where each stage can lose you cells, molecules or identity

Read left to right. The wet-lab stages are the same for both method families; what differs is only how cells are kept apart while their barcodes are written (Figs. 2, 4 and 5). Fragmentation is drawn explicitly because it is the step that decides whether a count means "molecules" or "molecules times length", and because ribosomal RNA depletion is placed relative to it (Fig. 7).

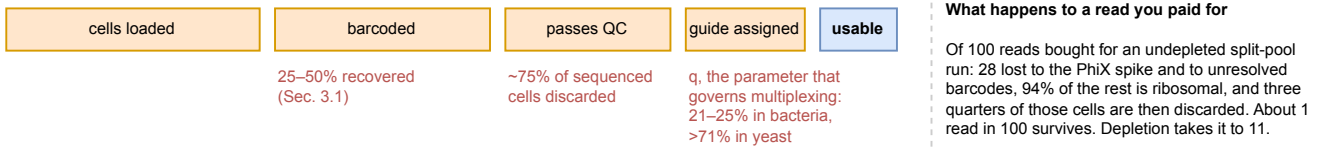
a The stages. Only the shaded block differs between split-pool and droplet.



b Fragmentation in bulk RNA-seq and in a 3'-counting single-cell assay. Same cutting, opposite consequence.



c What each stage costs you. The funnel, not the chemistry, is what a budget is made of.



Segments transcript / cDNA | barcode, and losses | enzyme or action | what you get | handle / adapter | H = PCR handle, one end only | CB = cell barcode

Panel b is the reason this figure exists. Bulk and single-cell protocols perform the SAME physical cutting, and the consequence inverts because of what happens afterwards: in bulk every piece can be sequenced, so reads scale with length; in a 3'-counting assay only the piece carrying the outer handle can be, so exactly one survives per molecule regardless of length. Applying TPM or FPKM to a 10x or SPLIT-seq count matrix is therefore an error, not a stylistic choice.